ModelIt!

Teacher's Guide

Is Social Media Rewiring Your Brain?

Modeling Dopamine Pathways, Attention Regulation, and Digital Addiction Mechanisms

Quick Reference

Grade Level	Grade 12L1
NGSS Standard	HS-LS1-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

© 2026 Discovery Collective. All rights reserved. For teacher use only.

Table of Contents

ModelIt!

1 NGSS Standards Unpacking & CAST Alignment 3

2 Scoring Rubric 4

3 Background Content Knowledge 5

4 The LEVER Framework 6

5 Slide-by-Slide Facilitation Guide 7

6 Answer Key & Expected Responses 9

7 Collaborative Learning Protocols 10

8 Differentiation & Extensions 11

9 Common Misconceptions 12

1 NGSS Standards & CAST Alignment

ModelIt!

Performance Expectation: HS-LS1-2

Develop and use a model to illustrate the hierarchical organization of interacting systems that provide specific functions within multicellular organisms.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop computational models to illustrate how hierarchically organized neural systems — from individual dopamine neurons to reward circuits to prefrontal cortex regulation — are disrupted by social media's designed reinforcement patterns.
Disciplinary Core Idea	**LS1.A Structure and Function** — The brain's reward circuitry (ventral tegmental area, nucleus accumbens, prefrontal cortex) provides specific motivational and attention-regulation functions that are altered by chronic social media stimulation patterns, affecting the hierarchical organization of cognitive control.
Crosscutting Concept	**Cause and Effect** — Students identify causal mechanisms linking specific social media design features (variable-ratio reinforcement, notification systems) to measurable changes in dopamine sensitivity and attention capacity through neuroplastic reorganization.

CAST Testing Alignment

MULTIPLE CHOICE:

A student spends 4 hours daily on TikTok for 3 months, then tries to study for the SAT. Based on the social media-brain model, which neurological changes best explain why sustained focus on test preparation feels nearly impossible?

CONSTRUCTED RESPONSE:

Using your model, compare the predicted neurological effects of 3 hours of passive Instagram scrolling per day versus 3 hours of active content creation (making art, writing, filming). Explain why the same total screen time produces different effects on Dopamine Sensitivity and Attention Capacity.

2 Scoring Rubric

ModelIt!

Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

ModelIt!

For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Silicon Valley's most profitable products are designed using the same behavioral neuroscience that makes slot machines addictive. Former tech insiders — including Facebook's founding president Sean Parker, Google's former design ethicist Tristan Harris, and the creators of the iPhone's infinite scroll — have publicly warned that social media platforms are engineered to exploit dopamine-driven feedback loops in the human brain. The question is no longer whether social media affects the brain, but how much damage is being done and whether it's reversible.

The Neuroscience of the Scroll

Social media platforms use variable-ratio reinforcement — delivering rewards (likes, comments, interesting content) at unpredictable intervals. This is the most addictive reinforcement schedule known to behavioral science, identical to the mechanism that makes slot machines the most profitable form of gambling. Each scroll is a pull of the lever: sometimes you get nothing interesting, sometimes you get a dopamine-triggering post. The unpredictability is the key — the brain releases more dopamine in anticipation of an uncertain reward than in response to a guaranteed one. This is why you can't stop scrolling even when most content is boring.

Dopamine Downregulation and Anhedonia

When the brain is chronically flooded with dopamine from social media stimulation, it protects itself by reducing dopamine receptor density — the same downregulation mechanism seen in substance addiction. The result is a lowered dopamine baseline: the brain's resting state of pleasure and motivation drops below normal. Activities that once felt rewarding — reading, face-to-face conversation, walking in nature — no longer produce enough dopamine to feel satisfying. Stanford psychiatrist Anna Lembke calls this 'dopamine deficit state' — a mild but pervasive anhedonia (inability to feel pleasure) that drives the user back to the screen for relief.

Attention Fragmentation and Neuroplasticity

The average TikTok video is 15-60 seconds. Instagram Reels, YouTube Shorts, and Twitter/X are all converging toward shorter content because engagement data shows users prefer rapid novelty. This preference isn't innate — it's trained. Neuroplasticity means the brain strengthens whatever neural circuits are used most frequently. Hours of daily rapid-content-switching strengthens circuits for quick task-switching while weakening circuits for sustained attention. Research by Gloria Mark at UC Irvine shows average attention span on a single screen has decreased from 2.5 minutes in 2004 to 47 seconds in 2023 — and heavy social media users show the shortest spans.

The Adolescent Vulnerability Window

The adolescent brain is uniquely vulnerable because the prefrontal cortex — responsible for impulse control, long-term planning, and self-regulation — doesn't fully mature until age 25. Meanwhile, the reward circuitry (nucleus accumbens) is fully active and hypersensitive during adolescence. This creates a developmental mismatch: teens have a fully operational gas pedal (reward-seeking) with an underdeveloped brake (impulse control). Social media exploits this window — which is why teen usage rates, addiction indicators, and mental health impacts consistently exceed those of adults exposed to identical platforms.

4 The LEVER Framework

ModelIt!

About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify social media exposure, notification frequency, dopamine sensitivity, attention capacity, and cognitive well-being as the key components of the social media-brain system.

E **Establish Relationships**
Students determine that variable-ratio reinforcement and notification interruptions drive dopamine dysregulation, that chronic exposure reduces dopamine sensitivity and attention capacity through neuroplasticity, and that usage pattern (passive vs. active) affects outcomes differently.

V **Visualize & Model**
Students build a computational model showing how social media design features interact with the brain's dopamine and attention systems to produce measurable changes in cognitive function over time.

E **Evaluate Outcomes**
Students run heavy passive use, mindful active use, and digital detox scenarios to test predictions about how different usage patterns affect dopamine sensitivity, attention capacity, and cognitive well-being.

R **Revise & Extend**
Students add social comparison intensity, sleep disruption, or FOMO activation to explore how additional psychological mechanisms interact with the core dopamine and attention model.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

ModelIt!

Slide 1: Cover

2 min

Say: *"You check your phone 96 times a day. That's not a personal choice — it's a design specification. Today we investigate why."*

Watch For

Show cover image. Ask: Estimate your daily screen time. Then check your phone's actual report. Surprised?

Slide 2: Learning Objectives

3 min

Say: *"Here's what you'll understand by the end — the neuroscience behind why you can't put your phone down."*

Watch For

Have students read objectives aloud. Pre-teach vocabulary with reward circuit diagram showing dopamine pathways.

Slide 3: Big Question

3 min

Say: *"Social media was designed by neuroscientists to be as addictive as possible. Is it physically changing your brain?"*

Watch For

Quick-write: Students write honestly about their own social media habits and any effects they've noticed. Save to revisit.

Slide 4: LEVER Framework

2 min

Say: *"We're going to model how the same dopamine system targeted by addictive substances is being targeted by your phone."*

Watch For

Preview each LEVER step. Emphasize that this is personal — the model describes what's happening in THEIR brains right now.

Slide 5: Activity 1: Components

8 min

Say: *"What are the key parts of this system? What do the platforms control, and what happens inside your brain?"*

Watch For

Guide sorting of external vs. internal components. Discuss: What's designed by the platform vs. what's your brain's response?

Slide 6: Activity 2: Connections

8 min

Say: *"After 3 hours of TikTok, you try to read a textbook. What does your model predict about Attention Capacity? Why does reading feel impossible?"*

Watch For

Students draw arrows showing +/- relationships. Discuss the dopamine downregulation cycle and attention fragmentation.

Slide 7: Activity 3: Simulation

10 min

Say: *"Run each scenario. Compare heavy passive scrolling, mindful active use, and digital detox. The differences are dramatic."*

Watch For

Students run all scenarios, record predictions vs. observations. Discuss what recovery actually looks like neurologically.

Slide 8: Discoveries

5 min

Say: *"The science is clear: social media exploits your dopamine system by design, and your brain physically changes in response. But recovery IS possible."*

Watch For

Class discussion of discoveries. Revisit initial reflections — has the model changed how you think about your own usage?

Slide 9: STEM Challenge

12 min

Say: *"Now use the same neuroscience that makes social media addictive — but in reverse. Design an app that builds healthy digital habits."*

Watch For

Groups design wellness apps using behavioral neuroscience principles from the model. Present and evaluate designs.

6 Answer Key & Expected Responses

ModelIt!

Component Sorting

- **EXTERNAL:** Social Media Exposure, Notification Frequency
- **INTERNAL:** Dopamine Sensitivity, Attention Capacity, Cognitive Well-Being

Relationships

- **Social Media Exposure to Dopamine Sensitivity:** NEGATIVE (-) — Chronic high social media exposure floods the reward circuit with dopamine, triggering receptor downregulation that lowers baseline dopamine sensitivity over time.
- **Notification Frequency to Attention Capacity:** NEGATIVE (-) — Frequent notifications train the brain to expect and seek constant interruptions, weakening the neural circuits for sustained attention through neuroplastic reorganization.
- **Dopamine Sensitivity to Cognitive Well-Being:** POSITIVE (+) — Higher dopamine sensitivity means normal activities produce adequate pleasure and motivation, supporting healthy mood, engagement, and executive function.
- **Attention Capacity to Cognitive Well-Being:** POSITIVE (+) — Greater attention capacity supports deeper learning, better emotional regulation, and more effective problem-solving — all core components of cognitive well-being.

Key Discoveries

- Social media platforms use variable-ratio reinforcement — the same mechanism that makes slot machines addictive — to create unpredictable reward patterns that maximize dopamine release and compulsive checking behavior
- Chronic heavy use lowers dopamine baseline sensitivity through receptor downregulation, making non-digital activities (reading, conversation, nature) feel increasingly boring and unrewarding
- Attention fragmentation is a neuroplastic change — the brain physically strengthens rapid-switching circuits and weakens sustained-attention circuits, measurably reducing the ability to focus on tasks longer than a few minutes
- Recovery is possible but not instant — dopamine sensitivity and attention capacity can partially restore during digital detox, but full recovery requires weeks to months depending on the duration and intensity of prior use

7 Collaborative Learning Protocols

ModelIt!

Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

ModelIt!

Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Social media is just a tool — it's how you use it that matters"	While usage patterns do affect outcomes (passive scrolling is worse than active creation), the platforms themselves are engineered to exploit neurological vulnerabilities. Variable-ratio reinforcement, infinite scroll, autoplay, algorithmic content curation, and notification systems are deliberate design choices made by behavioral scientists to maximize time-on-platform. Saying social media is 'just a tool' ignores that the tool was specifically designed to be addictive. A hammer is just a tool. A slot machine is a tool designed to exploit dopamine pathways — and so is social media.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"I can stop anytime I want — I'm not addicted"	The model shows that dopamine downregulation happens gradually and below conscious awareness. Most people don't recognize their baseline has shifted until they try to stop — and experience the restlessness, boredom, and irritability of dopamine deficit. Research shows that 50% of teens describe themselves as 'addicted' to their phones, and most who attempt digital detox fail within the first 72 hours. The hallmark of addiction is continued use despite negative consequences and difficulty stopping despite wanting to.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Attention spans aren't really getting shorter — that's just older people complaining"	Measurable data contradicts this. Gloria Mark's research at UC Irvine tracked attention span on screens from 2004-2023, showing a decline from 2.5 minutes to 47 seconds. Standardized cognitive testing shows declining sustained attention scores in heavy social media users. Brain imaging studies show reduced gray matter volume in the prefrontal cortex of heavy users — the same region that controls sustained attention. This isn't generational bias — it's neuroscience measured with fMRI machines and standardized tests.